

Apache SystemML: A Machine Learning Platform Suited for Big Data

Apache SystemML is an important machine learning platform that focuses on Big Data, with scalability and flexibility as its strong points. Its unique characteristics include algorithm customisation, multiple execution modes and automatic optimisation. This article introduces readers to the core features of Apache SystemML.



Machine learning (ML) has applications across various domains, and has transformed the manner in which these are built. The traditional sequential algorithmic approaches are now getting replaced with learning based dynamic algorithms. ML's most important benefit is its ability to handle novel scenarios.

Machine learning research can be divided into two parts — one is the development of the underlying ML algorithms, which requires a detailed understanding of core mathematical concepts. The other part is the application of machine learning algorithms, which doesn't require the developer to know the underlying mathematics down to the smallest detail. The second part, i.e., the application of ML, involves people from various domains. For example, ML is now applied in bio-informatics, economics, earth sciences, etc.

Another positive change in the ML space is the creation of various frameworks and libraries by many leading IT majors. These frameworks have made both the development and the application of ML easier and more efficient. As of

2019, developers no longer need to burden themselves with the implementation of core components. Most of these components are available as off-the-shelf solutions.

This article explores an important machine learning platform from Apache called SystemML, which focuses on Big Data. The sheer volume and velocity of Big Data poses the challenge of scalability. One of the important advantages of Apache SystemML is its ability to handle these scalability problems. The other distinguishing features of Apache SystemML (Figure 2) are:

- The ability to customise algorithms with the help of R-like and Python-like programming languages.
- The ability to work in multiple execution modes, which incorporate Spark MLContext, Spark Batch, etc.
- The ability to do optimisation automatically, which is based on the characteristics of both the data and cluster.

Apache SystemML has various components, all of which cannot be covered in this article. Here, we provide an introduction to the core features of Apache SystemML.